

two key issues: 1) A lack of interpretability, as their decisions, rooted in complex computations and high-dimensional representations, are challenging to articulate [4]. 2) They struggle to fully leverage textual financial information due to the high-dimensional nature and computational intensity of rich text embeddings [11, 13]. Consequently, DRL agents often rely on extracting textual sentiment [39], sidestepping the direct use of embeddings [6, 3], leading to an incomplete representation of crucial market information embedded in news and macroeconomic policies.

2.2 Advancements from LLMs to LLM Autonomous Agents

The evolution of LLMs has reshaped artificial intelligence and natural language processing. From foundational embeddings like Word2Vec [16] and GloVe [38], the field advanced with the introduction of sequential modeling like Long Short-Term Memory (LSTM) [18] and early transformer models of Bidirectional Encoder Representations from Transformers (BERT) [11]. Today, the new-generation LLMs, like Generative Pre-trained Transformer series (GPTs) [40, 34] and LLM Meta AI (Llamas) [48], stand out in diverse QA tasks. The trend leans towards LLM agents. While LLM agents for domain-specific tasks have been extensively researched [20, 25, 36], their application in financial trading remains underexplored. Existing studies in this domain, such as [52, 55], often lack open-source availability or have not considered an architecture specifically tailored to fit the unique environment of finance markets. Thus, there’s significant value in further investigating advanced, transparent LLM agents for trading.

2.3 Architecture Design of LLM Autonomous Agent

As Wang et al.[50] emphasizes, an effective architecture for LLMs serving autonomous agents is essential. Typically, this structure comprises modules like profiling, memory, planning, and actions, though not all may be essential for every application. There are cases of two modules (e.g., planning and action modules) being integrated as one component ([35]). The design variations are numerous. For instance, profiling has been achieved through methods like handcrafting [57], generation by LLMs [51], and alignment with real-world datasets [2]. Among these modules, the memory component is essential. Acting as the operational core, it aligns an agent’s actions with real-world tasks. Research indicates that leveraging insights from cognitive science studies on human memory [50, 45] can enhance this alignment. Thus, a well-structured trading LLM agent, comprising aptly designed modules, can sharply tackle the complexities of financial markets to make informed decisions.

3 Architecture of FINMEM

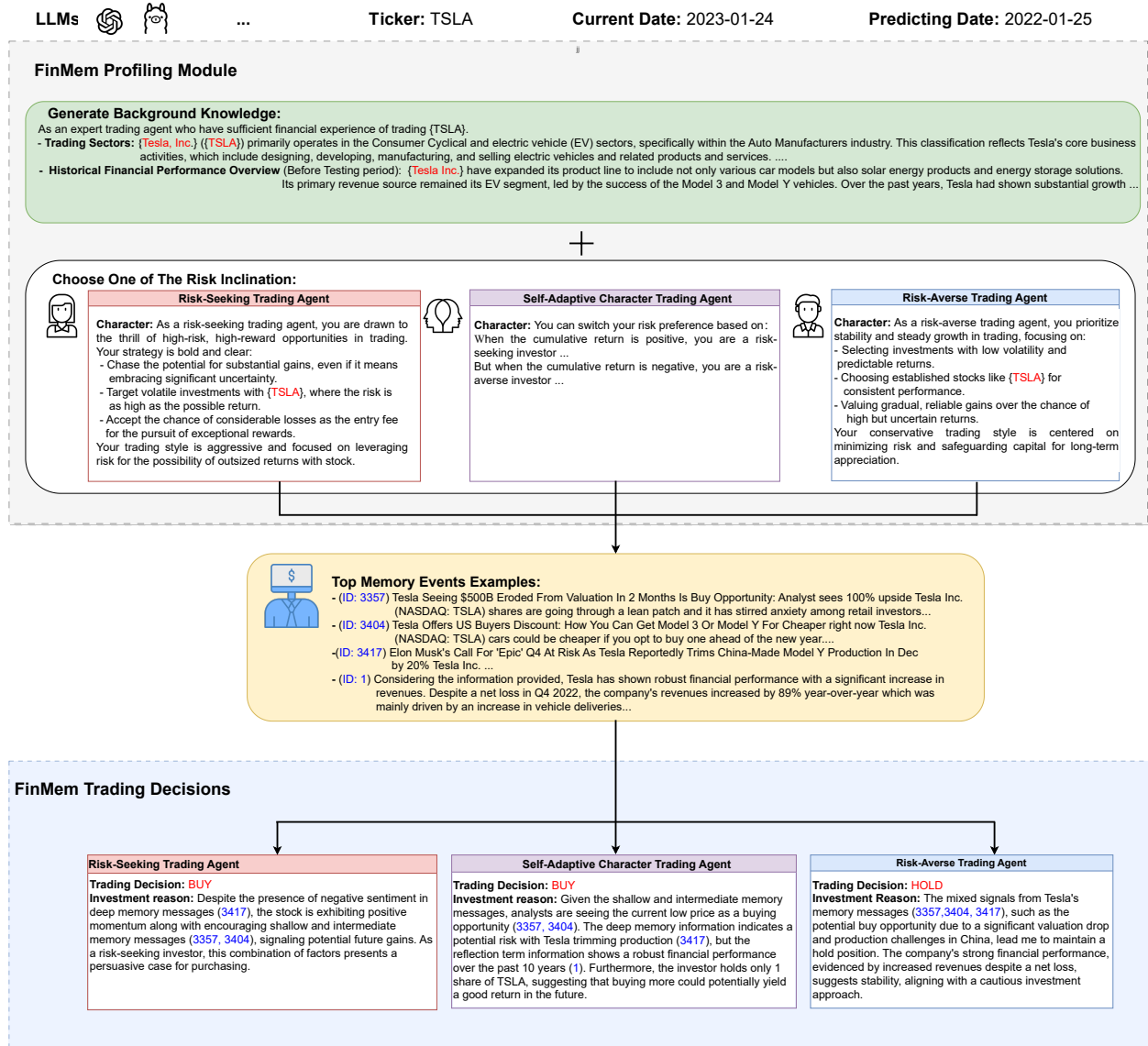
In this section, we comprehensively detail the three core components of FINMEM, namely the profiling, memory, and decision-making modules. The profiling module empowers FINMEM to adaptively tailor character setting for specific trading tasks. Memory module leverages diverse time-efficiency attributes of financial data, enhancing its trading efficacy. The decision-making module enables FINMEM to synchronize its memory streams with market facts, facilitating high-quality trading decisions. The details and notations associated with these three modules are provided in the subsequent sections.

3.1 Profiling Module

The profiling module empowers FINMEM to develop a dynamic agent character specifically designed to navigate the complex dynamics of financial markets effectively.

The dynamic character of FINMEM comprises two principal components, as depicted in Figure 1: firstly, a foundational professional knowledge base akin to a trading expert, and secondly, an agent with three distinct investment risk inclinations. The first component includes two types of information: an introduction to the primary trading sectors relevant to the company stock FINMEM will trade in, and a concise overview of the historical financial performance of the specified ticker, spanning from the beginning to the end of the training period. Before initiating trades in a new company’s stock, FINMEM accesses and updates this sector-specific and historical financial data from a backend database. This professional background setting narrows down information and memory events pertinent to specific trading tasks.

The second component of FINMEM’s design, illustrated in Figure 1, encompasses three distinct risk inclination options: risk-seeking, risk-averse, and a self-adaptive risk character. The risk-seeking setting gears FINMEM towards an aggressive, high-reward approach, while the risk-averse setting gears it towards a conservative, lower-risk strategy. A distinctive aspect of FINMEM is its ability to dynamically alternate between these risk settings in response to current market conditions. Specifically, it shifts risk preferences when the Cumulative Return falls to below zero within a brief period, such as three days, and reversely. This flexible design functions as a protective mechanism, mitigating prolonged



downturns in turbulent market environments. During the initial stage of the training phase, FINMEM is configured with a chosen risk preference, each supplemented with comprehensive textual explanations through LLM prompts. These guidelines shape how FINMEM processes incoming messages and determines its subsequent actions in alignment with its designated risk inclination. The system maintains a catalog of all risk inclinations and their detailed explanations in a backlog, enabling seamless adaptation to different stocks by switching among these risk profiles as needed.

The dynamic character setting in FINME’s profiling module provides subjective and professional background knowledge and flexible choice of risk inclinations. It provides crucial context for filtering and retrieving trading-relevant information and memory events, thus improving accurate inferencing and adaptability to fluctuating market conditions.

3.2 Memory Module

The memory module of FINMEM emulates a human trader’s cognitive system so that it can efficiently process hierarchical financial information and prioritize the critical messages for high-quality investment decisions. Furthermore, it adjusts the memory span flexibly, enabling the agent to operate on a wider range of events over a longer retrieval period. FINMEM’s memory module, illustrated in Figure 2, comprises working and long-term memory with layered processing capability and is initiated by a specific investment inquiry.

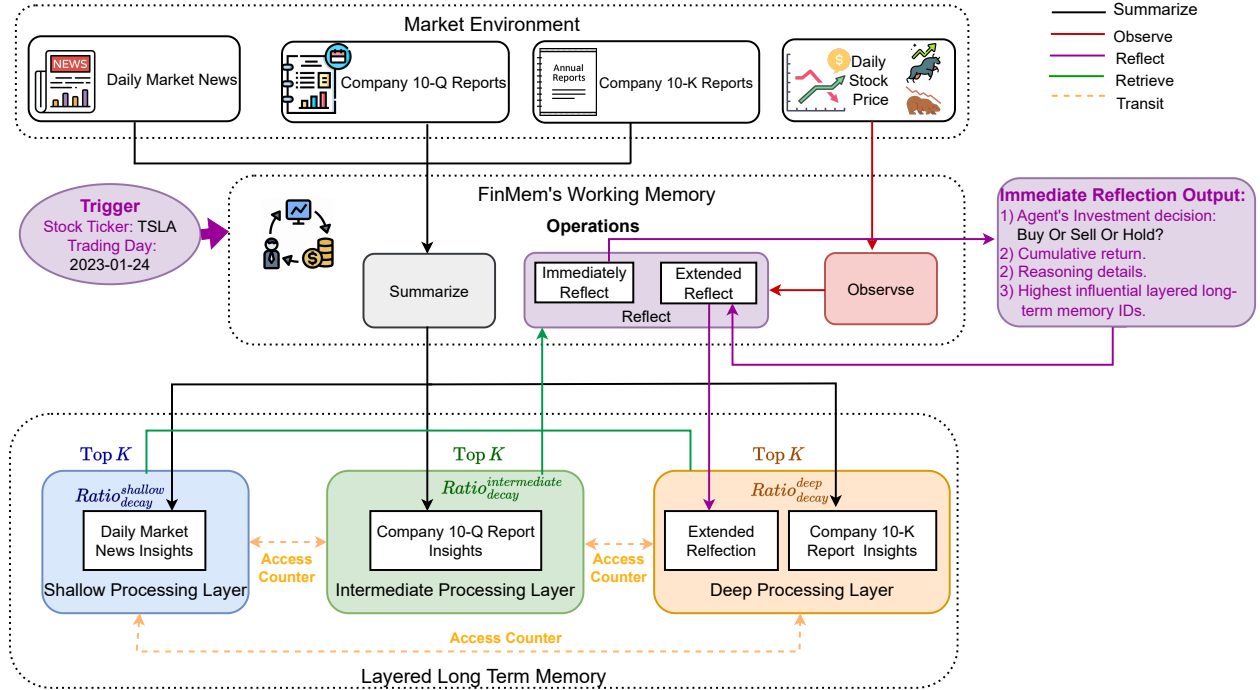


Figure 2: Memory module structure of FINMEM with a detailed view of components, operations, and workflow. The cognitive architectures of FINMEM’s memory module have two core components – Working Memory and Layered Long-term Memory.

3.2.1 Working memory

Working memory refers to the human cognitive system’s functions for temporary storage and diverse operations. We incorporate this concept into FINMEM’s memory module development, creating a central workspace for informed decision-making. Unlike human working memory, having a maximum capacity of seven plus or minus two memory events [30], FINMEM has the ability to expand the capacity based on specific requirements. Tailored for converting financial data into trading actions, FINMEM’s working memory encompasses three key operations: summarization, observation, and reflection. The mechanisms by which they interact and operate as an integrated decision-making workflow are detailed in the middle box of Figure 2. Additionally, the LLM prompt template that underpins these processes is thoroughly outlined in 3.

Summarization: FINMEM leverages external market data to derive critical investment insights and sentiments tailored to specific stock trading queries, such as “Can you make an investment decision on TSLA on 1/24/2023?”. As